

ROBERT VOKÁČ, BSc

C++ Systems Engineer · Backend Platforms · Framework Architecture
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PROFILE

Focused on **system-level development, backend platforms, and framework architecture** using modern C++ (C++20/23).

Author of a 40k+ LOC **modular backend platform and a real-time rendering framework**. Strong focus on **API design, abstraction layers, and extensible system architecture**.

Seeking roles in C++ systems engineering, backend/platform development, or framework/engine architecture.

KEY PROJECTS

Mind-Net - Modular C++ Backend Platform (~40k LOC)

GitHub: github.com/robertvokac/mind-net

Designed and implemented a **metadata-driven backend platform from scratch in modern C++**.

- Multi-module architecture with **plugin-based extensibility**
- **Service orchestration** layer coordinating **validation, triggers, and persistence**
- **HTTP layer with generated routing and request handling**
- Custom **ORM and data modeling layer**
- **REST API and database layer generation**
- Event-driven system with **Before/After/InsteadOf triggers**
- **Custom migration engine** with integrity checks
- Built-in **scheduler and background job system**

Focus: **backend systems, platform architecture, extensibility**

CNA - Real-Time Rendering Engine / C++ Framework

GitHub: github.com/openeggbert/cna

Cross-platform real-time rendering engine inspired by XNA, built in modern C++.

- Engine-style architecture with **clear API / backend separation**
- Rendering pipeline abstraction over SDL 3
- Resource management and system lifecycle design
- Focus on reusable framework components and portability

Focus: **low-level systems, rendering and runtime architecture**

easy-gl - Rendering Infrastructure Library

GitHub: github.com/openeggbert/easy-gl

- Lightweight OpenGL abstraction layer for reusable rendering support.

cpp-dotnet - C++ Foundation Layer

GitHub: github.com/openeggbert/cpp-dotnet

- Reusable system components (events, delegates, exceptions).

TECHNICAL SKILLS

Languages

C++20/23 (primary), SQL

Systems & Tools

CMake, Git, Linux, Bash, GCC/Clang, SDL 3, SQLite

Core Competencies

- System architecture and backend platforms
- Framework and engine design
- Modular systems and plugin architecture
- API design and abstraction layers
- Event-driven systems
- Custom ORM and migration tooling

PROFESSIONAL EXPERIENCE

(Note: current role does not involve C++; C++ expertise developed through independent system-level projects.)

Software Engineer (Backend Systems) - Assist, spol. s r. o.

2017 - present

- Worked on long-running backend systems with focus on maintainability, data modeling, and system stability.
- Applied system design principles in large, long-lived codebases
- SQL database design and optimization
- Contribution to system architecture and maintainability
- Team collaboration using Git and Agile workflows
- Long-term focus on system design, maintainability, and backend architecture

Technologies: Java, SQL, Git, Bash

EDUCATION

BSc in Computer Science - Czech University of Life Sciences Prague (2014-2017)

Secondary Technical School Hořovice — Informatics in Economics (2008-2012)

LANGUAGES

English — B2 (technical documentation, team communication)